

MATHEMATICS AND STATISTICS

TIME: 3 HOURS

[MAX. MARKS: 80]

- Note:** (i) All questions are compulsory.
(ii) Figures to the right indicate full marks.
(iii) Graph of L.P.P should be drawn on graph paper only.
(iv) Use of logarithmic table is allowed.
(v) Answers to the questions of Section – I and Section –II should be written in the only one answer book.
(vi) Answer to every new question must be written on a new page.

Question 1: The principle solution of $\cot x = -\sqrt{3}$

- (a) $\frac{\pi}{6}, \frac{5\pi}{6}$ (b) $\frac{5\pi}{6}, \frac{7\pi}{6}$
(c) $\frac{5\pi}{6}, \frac{11\pi}{6}$ (d) $\frac{\pi}{6}, \frac{11\pi}{6}$

Solution 1:

$$\cot x = -\sqrt{3}$$

$$\text{we know that } \cot 30^\circ = -\sqrt{3}$$

Here $\tan x$ is negative

$\tan$ is negative in 2nd and 4th quadrant

$$\text{Value in 2nd quadrant} = 360^\circ - 30^\circ = 330^\circ$$

So the principle solution are,

$$x = 30^\circ$$

$$x = 30 \times \frac{\pi}{180}$$

$$x = \frac{\pi}{6}$$

and,

$$x = 360^\circ$$

$$= 360 \times \frac{\pi}{180}$$

$$\frac{11\pi}{6}$$

Hence, the solution is $\frac{\pi}{6}, \frac{11\pi}{6}$.

Question 2: The acute angle between the two planes $x + y + 2z = 8$ and $3x + 2y + 2z = 7$ is.....

(a) $\sin^{-1}\left(\frac{5}{\sqrt{102}}\right)$

(b) $\cos^{-1}\left(\frac{5}{\sqrt{102}}\right)$

(c) $\sin^{-1}\left(\frac{15}{\sqrt{102}}\right)$

(d) $\cos^{-1}\left(\frac{15}{\sqrt{102}}\right)$

Solution 2:

Given that,

$$x + y + 2z = 8 \text{ and } 3x - 2y + 2z = 7$$

Now,

$$\vec{N}_1 = \hat{i} + \hat{j} + 2\hat{k}$$

$$\vec{N}_2 = 3\hat{i} - 2\hat{j} + 2\hat{k}$$

Angle between two vectors

$$\vec{N}_1 \cdot \vec{N}_2 = |\vec{N}_1| \cdot |\vec{N}_2| \cdot \cos \alpha$$

$$\cos \alpha = \frac{\vec{N}_1 \cdot \vec{N}_2}{|\vec{N}_1| \cdot |\vec{N}_2|}$$

$$\vec{N}_1 \cdot \vec{N}_2 = 1 \cdot 3 + 1 \cdot (-2) + 2 \cdot 2$$

$$= 3 - 2 + 4 = 5$$

$$|\vec{N}_1| = \sqrt{1^2 + 1^2 + 2^2}$$

$$= \sqrt{6}$$

$$|\vec{N}_2| = \sqrt{3^2 + (-2)^2 + 2^2}$$

$$= \sqrt{9 + 4 + 4}$$

$$= \sqrt{17}$$

$$\therefore \cos \alpha = \frac{5}{\sqrt{6} \cdot \sqrt{17}} = \frac{5}{\sqrt{102}}$$

$$\alpha = \cos^{-1} \frac{5}{\sqrt{102}}$$

Hence, correct option is (B).

Question 3: The direction ratios of the line which is perpendicular to the lines with direction ratios $-1, 2, 2$ and $0, 2, 1$ are.....

(a) $-2, -1, -2$

(b) $2, 1, 2$

(c) $2, -1, -2$

(d) $-2, 1, -2$

Solution 3:

Given the perpendicular to the line whose direction is $-1, 2, 2$

So,

$$\begin{aligned} a(-1) + b(2) + c(2) \\ -a + 2b + 2c = \dots\dots\dots (i) \end{aligned}$$

Again,

$$\begin{aligned} a(0) + b(2) + c(1) \\ 2b + c = \dots\dots\dots (ii) \end{aligned}$$

Now, the solving the equation (i) and (ii)

$$\begin{aligned} \left| \begin{array}{ccc} a & b & c \\ -1 & 2 & 2 \\ 0 & 2 & 1 \end{array} \right| \\ = \frac{a}{2-4} = \frac{b}{-1-0} = \frac{c}{-2-0} \\ = \frac{a}{-2} = \frac{b}{-1} = \frac{c}{-2} \end{aligned}$$

Hence, the correct option is (a).

Question 4: If $f(x) = (1+2x)^{\frac{1}{x}}$ for $x \neq 0$ is continuous at $x=0$ then, $f(0) = \dots\dots\dots$

- (a) e (b) e^2
(c) 0 (d) 2

Solution 4:

We have,

$$\begin{aligned} \lim_{x \rightarrow 0^+} f(x) &= \lim_{x \rightarrow 0^+} (1+2x)^{\frac{1}{x}} \\ &= \lim_{x \rightarrow 0^+} \left\{ (1+3x)^{\frac{1}{3x}} \right\}^3 \\ &= e^3 \end{aligned}$$

Question 5: $\int \frac{dx}{9x^2+1} = \dots\dots\dots$

$$(a) \frac{1}{3} \tan^{-1}(2x) + c$$

$$(b) \frac{1}{3} \tan^{-1} x + c$$

$$(c) \frac{1}{3} \tan^{-1}(3x) + c$$

$$(d) \frac{1}{3} \tan^{-1}(6x) + c$$

Solution 5:

Given that,

$$\begin{aligned} \frac{1}{9} \int \frac{dx}{x^2 + \frac{1}{9}} &= \frac{1}{9} \int \frac{dx}{x^2 + \left(\frac{1}{3}\right)^2} \\ &= \frac{1}{9} \times \frac{1}{\frac{1}{3}} \tan^{-1} \left(\frac{x}{\frac{1}{3}} \right) + c \\ &= \frac{1}{3} \tan^{-1}(3x) + C \end{aligned}$$

Question 6: If $y = ae^{5x} + be^{-5x}$, then the differential equation is.....

$$(a) \frac{d^2 y}{dx^2} = 25y$$

$$(b) \frac{d^2 y}{dx^2} = -25y$$

$$(c) \frac{d^2 y}{dx^2} = -5y$$

$$(d) \frac{d^2 y}{dx^2} = 5y$$

Solution 6:

Given that,

$$\begin{aligned} \therefore \frac{dy}{dx} &= 5 \cdot a \cdot e^{5x} + 5b \cdot e^{-5x} \\ \frac{d^2 y}{dx^2} &= 25ae^{5x} + 25be^{-5x} \\ &= 25(ae^{5x} + be^{-5x}) \\ &= \frac{d^2 y}{dx^2} = 25y \end{aligned}$$

Hence, the correct option is (a).

Question 7: Write the truth values of the following statement:

(i) 2 is a rational number and $\sqrt{2}$ is an irrational number.

(ii) $2+3=5$ or $\sqrt{2}+\sqrt{3}=\sqrt{5}$

Question 8: Find the volume of the parallelepiped if coterminous edges are given by the vectors $2\hat{i}+5\hat{j}-4\hat{k}$, $5\hat{i}+7\hat{j}+5\hat{k}$, $4\hat{i}+5\hat{j}-2\hat{k}$.

Solution 8:

Given that,

$$\vec{a} = 2\hat{i} + 5\hat{j} - 4\hat{k}$$

$$\vec{b} = 5\hat{i} + 7\hat{j} + 5\hat{k}$$

$$\vec{c} = 4\hat{i} + 5\hat{j} - 2\hat{k}$$

$$\begin{aligned} \vec{a} \cdot (\vec{b} \times \vec{c}) &= \begin{vmatrix} a_x & a_y & a_z \\ b_x & b_y & b_z \\ c_x & c_y & c_z \end{vmatrix} = \begin{vmatrix} 2 & 5 & -4 \\ 5 & 7 & 5 \\ 4 & 5 & -2 \end{vmatrix} \\ &= 2(-14 - 25) - (-10 - 20) - 4(25 - 28) \\ &= 2 \times (-39) + 150 + 12 \\ &= -78 + 150 + 12 \\ &= -78 + 162 \\ &= 84 \\ [\vec{a} \vec{b} \vec{c}] &= 84 \\ \vec{v} &= |([\vec{a} \vec{b} \vec{c}])| = |84| \\ &= 84 \end{aligned}$$

OR

Question 8: Find the value of p, if the vector $\hat{i} - 2\hat{j} + \hat{k}$, $2\hat{i} - 5\hat{j} + p\hat{k}$ and $4\hat{i} + 5\hat{j} - 2\hat{k}$ are coplanar.

Solution 8:

Since, $\vec{a}, \vec{b}, \vec{c}$ are coplanar vectors,

$$[\vec{a} \ \vec{b} \ \vec{c}] = 0$$

$$\therefore \begin{vmatrix} 1 & -2 & 1 \\ 2 & -5 & p \\ 4 & 5 & -2 \end{vmatrix} = 0$$

$$\therefore 1(10 - 5p) + 2(-4 - 4p) + 1(10 + 20) = 0$$

$$\Rightarrow 10 - 5p - 8 - 8p + 30 = 0$$

$$\Rightarrow 40 - 8 - 8p = 0$$

$$\Rightarrow 32 - 8p = 0$$

$$p = \frac{32}{8}$$

Question 9: Show that the points $A(-7, 4, -2)$, $B(-2, 1, 0)$ and $C(3, -2, 2)$ are collinear.

Solution 9: Let the position vector of points A, B and C be

$$\vec{OA} = -7\hat{i} + 4\hat{j} - 2\hat{k}$$

$$\vec{OB} = -2\hat{i} + \hat{j}$$

$$\vec{OC} = 3\hat{i} - 2\hat{j} + 2\hat{k}$$

Now,

$$\vec{AB} = \vec{OB} - \vec{OA}$$

$$= (-2\hat{i} + \hat{j}) - (-7\hat{i} + 4\hat{j} - 2\hat{k})$$

$$= 5\hat{i} - 3\hat{j} + 2\hat{k}$$

$$\vec{BC} = \vec{OC} - \vec{OB}$$

$$= (3\hat{i} - 2\hat{j} + 2\hat{k}) - (-2\hat{i} + \hat{j})$$

$$= 5\hat{i} - 3\hat{j} + 2\hat{k}$$

$$\begin{aligned}\vec{AC} &= \vec{OC} - \vec{OA} \\ &= (3\hat{i} - 2\hat{j} + 2\hat{k}) - (-7\hat{i} + 4\hat{j} - 2\hat{k}) \\ &= 10\hat{i} - 6\hat{j} + 4\hat{k} \\ &= 2(5\hat{i} - 3\hat{j} + 2\hat{k})\end{aligned}$$

$$\begin{aligned}|\vec{AB}| &= \sqrt{5^2 - 3^2 + 2^2} \\ &= \sqrt{25 - 9 + 4} \\ &= \sqrt{20}\end{aligned}$$

$$\begin{aligned}|\vec{BC}| &= \sqrt{5^2 - 3^2 + 2^2} \\ &= \sqrt{25 - 9 + 4} \\ &= \sqrt{20}\end{aligned}$$

$$\begin{aligned}|\vec{AC}| &= \sqrt{10^2 - 6^2 + 4^2} \\ &= \sqrt{100 - 36 + 16} \\ &= \sqrt{80} \\ &= 2\sqrt{20}\end{aligned}$$

$$\text{Hence, } AB = \sqrt{20}, BC = \sqrt{20}, AC = 2\sqrt{20}$$

Thus,

$$AB + BC = AC$$

Hence, A, B, C are collinear.

Question 10: Write the equation of the plane $3x + 4y - 2z = 5$ in the vector form.

Question 11: If $y = x^x$, find $y = x^x$

Solution 11: We have,

$$y = x^x$$

Then, taking log on both sides, we get,

$$\log(y) = \log(x^x)$$

$$\frac{d}{dx}(\log y) = x \log x$$

$$\frac{1}{y} \cdot \frac{dy}{dx} = x \cdot \frac{1}{x} + \log(x) \cdot 1$$

$$y \cdot \frac{1}{y} \frac{dy}{dx} = y(1 + \log(x))$$

$$\text{Thus, } \frac{dy}{dx} = x^x(1 + \log x)$$

Question 12: Find the equation of tangent to the curve $y = x^2 + 4x + 1$ at point $(-1, -2)$

Solution 12:

We have,

$$y = x^2 + 4x + 1$$

$$\therefore \frac{dy}{dx} = 2x + 4$$

$$\left(\frac{dy}{dx} \right)_{(-1, -2)} = -2 + 4$$

$$\therefore m = 2$$

Equation of tangent

$$y = mx + c \text{ ----- (1)}$$

Putting the point $(-1, -2)$ in equation (1)

$$-2 = 2 \times -1 + c$$

$$\Rightarrow -2 = -2 + c$$

$$\Rightarrow -2 + 2 = c$$

$$c = 0$$

$$\therefore y = 2x + 0$$

Question 13: Evaluate: $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$

Solution 13:

We have,

$$\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$$

Put $ex^x = z$

$$(ex^x + e^x)dx = dz$$

$$\text{or, } (1+x)e^x dx = dz$$

$$= \int \frac{dz}{\cos^2 z}$$

$$= \int \tan^2 z dz$$

$$= \tan z + c$$

$$= \tan(xe^x) + c$$

Question 14: Evaluate : $\int_0^{\frac{\pi}{2}} \sin^2 x dx$

Solution 14:

We have $\int_0^{\frac{\pi}{2}} \sin^2 x dx$

Now,

$$\int_0^{\frac{\pi}{2}} \frac{(1 - \cos 2x)}{2} dx$$

$$= \frac{1}{2} \int_0^{\frac{\pi}{2}} dx - \frac{1}{2} \int_0^{\frac{\pi}{2}} \cos 2x dx$$

$$= \frac{1}{2} [x]_0^{\frac{\pi}{2}} - \left[\frac{1}{4} \sin 2x \right]_0^{\frac{\pi}{2}} + C$$

$$= \frac{1}{2} \left[\frac{\pi}{2} \right] - \frac{1}{4} \left[\sin \frac{\pi}{2} \right] + C$$

$$= \left(\frac{\pi}{4} - 1 \right) + C$$

$$= \frac{\pi}{4} - \frac{1}{4} + C$$

$$= \frac{\pi - 1}{4} + C$$

Question 15: In $\triangle ABC$, prove that

$$\sin\left(\frac{B-C}{2}\right) = \left(\frac{b-c}{-a}\right) \cos\left(\frac{A}{2}\right)$$

OR,

Show that $\sin^{-1}\left(\frac{5}{13}\right) + \cos^{-1}\left(\frac{3}{5}\right) = \tan^{-1}\left(\frac{63}{16}\right)$

Question 16: If $A(\vec{a})$ and $B(\vec{b})$ are any two points in the space and $R(\vec{r})$ be a point on the line segment AB dividing it internally in the ratio $m:n$, then prove that

$$\vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

Solution 16:

Let O be the origin,

Then,

$$\vec{OA} = \vec{a} \text{ and } \vec{OB} = \vec{b}$$

Let P be a point on $\vec{AB}$

Such that,

$$\frac{\vec{AP}}{\vec{PB}} = \frac{m}{n}$$

$$\Rightarrow n(\vec{AP}) = m(\vec{PB})$$

$$\Rightarrow n(\vec{OP} - \vec{OA}) = m(\vec{OB} - \vec{OP})$$

$$\Rightarrow (m+n)\vec{OP} = m\vec{OB} + n\vec{OA}$$

$$\Rightarrow (m+n)\vec{OP} = m\vec{b} + n\vec{a}$$

$$\vec{OP} = \vec{r} = \frac{m\vec{b} + n\vec{a}}{m+n}$$

Hence, proved.

Question 17: The equation of a line is $2x - 2 = 3y + 1 = 6z - 2$, find its direction ratios and also find the vector equation of the line.

Solution 17:

Given that,

$$2x - 2 = 3y + 1 = 6z - 2$$

$$2(x - 1) = 3\left(y + \frac{1}{3}\right) = 6\left(z - \frac{1}{3}\right)$$

$$\frac{x-1}{3} = \frac{y+\frac{1}{3}}{2} = \frac{z-\frac{1}{3}}{1}$$

$$\vec{a} = \hat{i} + \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}$$

$$\vec{b} = 3\hat{i} + 2\hat{j} + \hat{k}$$

Vector equation of the line

$$\vec{r} = \left(\hat{i} + \frac{1}{3}\hat{j} - \frac{1}{3}\hat{k}\right) + \lambda(3\hat{i} + 2\hat{j} + \hat{k})$$

Question 18: Discuss the continuity of the function $f(x) = \frac{\log(2+x) - \log(2-x)}{\tan x}$, for $x \neq 0$
 $= 1$

for $x = 0$ at the point $x = 0$

Solution 18:

Given that,

$$\begin{aligned}
 \lim_{x \rightarrow 0} f(x) &= \lim_{x \rightarrow 0} \frac{\log(2+x) - \log(2-x)}{\tan x} \\
 &= \lim_{x \rightarrow 0} \frac{\log\left(\frac{2+x}{2-x}\right)}{\tan x} \\
 &= \lim_{x \rightarrow 0} \frac{\log\left(\frac{1+\frac{x}{2}}{1-\frac{x}{2}}\right)}{\tan x} \\
 &= \lim_{x \rightarrow 0} \frac{\lim_{x \rightarrow 0} \left(1+\frac{x}{2}\right) - \log\left(1-\frac{x}{2}\right)}{\tan x} \\
 &= \lim_{x \rightarrow 0} \frac{\lim_{x \rightarrow 0} \left(1+\frac{x}{2}\right) - \log\left(1-\frac{x}{2}\right)}{x} \\
 &= \lim_{x \rightarrow 0} \frac{\lim_{x \rightarrow 0} \frac{\log\left(1+\frac{x}{2}\right)}{\left(\frac{x}{2}\right) \times 2} + \lim_{x \rightarrow 0} \frac{\log\left(1-\frac{x}{2}\right)}{\left(-\frac{x}{2}\right) \times 2}}{\lim_{x \rightarrow 0} \frac{\tan x}{x}} \\
 &= \lim_{x \rightarrow 0} \frac{\frac{1}{2} \lim_{x \rightarrow 0} \frac{\log\left(1+\frac{x}{2}\right)}{\left(\frac{x}{2}\right)} + \frac{1}{2} \lim_{x \rightarrow 0} \frac{\log\left(1-\frac{x}{2}\right)}{\left(-\frac{x}{2}\right)}}{\lim_{x \rightarrow 0} \frac{\tan x}{x}} \\
 &= \frac{\frac{1}{2}(1) + \frac{1}{2}(1)}{1} \quad \left[\because \lim_{x \rightarrow 0} \frac{\log(1+x)}{x} = 1 \text{ and } \lim_{x \rightarrow 0} \frac{\tan x}{x} = 1 \right] \\
 &= 1
 \end{aligned}$$

Since

$$\lim_{x \rightarrow 0} f(x) = f(0), f \text{ is continuous at } x = 0.$$

Question 19: The probability distribution of a random variable X, the number of defect per 10 meters of a fabric is given by

X	0	1	2	3	4
P(X= x)	0.45	0.35	0.15	0.03	0.02

Find the variance of X.

Solution 19:

$$\begin{aligned}
 E(X) &= \sum_{i=1}^n x_i P(x_i) \\
 &= 0(0.45) + 1(0.35) + 2(0.15) + 3(0.03) + 4(0.02) \\
 &= 0 + 0.35 + 0.30 + 0.09 + 0.08 \\
 &= 0.82
 \end{aligned}$$

$$\begin{aligned}
 E(X^2) &= \sum_{i=1}^n x_i^2 P(x_i) \\
 &= 0(0.45) + 1(0.35) + 4(0.15) + 9(0.03) + 16(0.02) \\
 &= 0 + 0.35 + 0.60 + 0.27 + 0.32 \\
 &= 1.54
 \end{aligned}$$

$$\begin{aligned}
 \therefore \text{var}(X) &= E(X^2) - [E(X)]^2 \\
 &= 1.54 - (0.82)^2 \\
 &= 1.54 - 0.6724 \\
 &= 0.8676 \\
 \therefore \text{var}(X) &= 0.8676
 \end{aligned}$$

OR

Question 19: For the following probability density function (p.d.f) of X, find :

$$P(X < 1), (ii) P(|X| < 1)$$

$$\text{if } f(x) = \frac{x^2}{18}, -3 < x < 3$$

$$= 0, \text{ otherwise}$$

Solution 19:

$$\begin{aligned}
 P(X < 1) &= \int_{-3}^1 f(x) dx \\
 &= \int_{-3}^1 \frac{x^2}{18} dx \\
 &= \frac{1}{18} \left[\frac{x^3}{3} \right]_{-3}^1 \\
 &= \frac{1}{18} \left(\frac{1}{3} + \frac{27}{3} \right) \\
 &= \frac{28}{54} = \frac{14}{27} \\
 &= 0.5185 \\
 P(|X| < 1) &= P(-1 < X < 1) \\
 &= \int_{-1}^1 f(x) dx \\
 &= \int_{-1}^1 \frac{x^2}{18} dx \\
 &= \frac{1}{18} \left[\frac{x^3}{3} \right]_{-1}^1 \\
 \text{(i)} &= \frac{1}{18} \left(\frac{1}{3} + \frac{1}{3} \right) \\
 &= \frac{1}{27} \\
 &= 0.03703
 \end{aligned}$$

Question 20: Given is $X \sim B(n, p)$.

if $E(X) = 6, \text{Var.}(X) = 4 \cdot 2$ find the value of n .

Solution 20:

Given that,

$$X \sim B(n, p), E(X) = 6 \text{ and } \text{var}(X) = 4.2$$

But,

$$E(X) = np$$

$$\therefore 6 = np \text{-----(i)}$$

and,

$$\text{var}(X) = npq$$

$$4.2 = npq \text{-----(ii)}$$

$$4.2 = 6.q \quad [\text{from(i)}]$$

$$\therefore q = \frac{4.2}{6} = \frac{7}{10}$$

We know that,

$$p + q = 1$$

$$\therefore p = 1 - q$$

$$\therefore p = 1 - \frac{7}{10}$$

$$\therefore p = \frac{3}{10}$$

Putting the value of p in (i), we get

$$6 = n \times \frac{3}{10}$$

$$\Rightarrow n = \frac{6 \times 10}{3}$$

$$= 20$$

Hence, the value of $n = 20$

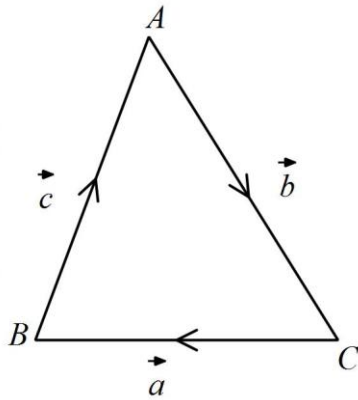
Question 21: Find the symbolic form of the given switching circuit. Construct its switching table and interpret your result.(figher)

Question 22: If three numbers are added, their sum is 2. If two times the second number is subtracted from the sum of first and third numbers we get 8 and if three times the first number is added to the sum of second and third numbers we get 4. Find the numbers using matrices.

Question 23: In any $\triangle ABC$, with usual notations, prove that $b^2 = c^2 + a^2 - 2ca \cos B$. (4)

Solution 23:

Let be $\triangle ABC$



In $\triangle ABC$,

$$\overrightarrow{BC} = \vec{a}, \overrightarrow{CA} = \vec{b}$$

$$\overrightarrow{BA} = \vec{c}$$

By law of sum of Δ 's

$$\vec{a} + \vec{b} = \vec{c}$$

$$\vec{b} = \vec{c} - \vec{a}$$

$$\vec{b} \cdot \vec{b} = \vec{b} \cdot (\vec{c} - \vec{a})$$

$$b^2 = (\vec{c} - \vec{a}) \cdot (\vec{c} - \vec{a})$$

$$b^2 = c^2 + a^2 - 2\vec{a} \cdot \vec{c}$$

$$\text{But, } \vec{a} \cdot \vec{c} = ac \cos B$$

$$\therefore b^2 = c^2 + a^2 - 2ca \cos B$$

OR,

Question 23: In $\triangle ABC$ with the usual notations prove that

$$(a-b)^2 \cos^2 \left(\frac{C}{2} \right) + (a+b)^2 \sin^2 \left(\frac{C}{2} \right) = c^2 \quad (4)$$

Solution 23:

By the projection rule in ΔABC

$$a = c \cos B + b \cos C$$

$$b = a \cos C + c \cos A$$

$$a + b = c \cos B + b \cos C + a \cos C + c \cos A$$

$$(a + b)(1 - \cos C) = c(\cos B + \cos A)$$

$$(a + b) \left(2 \sin^2 \left(\frac{C}{2} \right) \right) = c(\cos B + \cos A)$$

$$(a + b)^2 \cdot \sin^2 \left(\frac{C}{2} \right) = \frac{c}{2} (\cos B + \cos A)(a + b) \dots \dots \dots (1)$$

And,

$$a - b = c \cos B + b \cos C - a \cos C - c \cos A$$

$$(a - b)(1 + \cos C) = c(\cos B - \cos A)$$

$$(a - b) \left(\cos^2 \left(\frac{C}{2} \right) \right) = c(\cos B - \cos A)$$

$$(a - b)^2 \cos^2 \left(\frac{C}{2} \right) = \frac{c}{2} (\cos B - \cos A)(a - b) \dots \dots \dots (2)$$

Now, adding the equations (1) and (2), we get

$$\begin{aligned} & (a - b)^2 \cos^2 \left(\frac{C}{2} \right) + (a + b)^2 \sin^2 \left(\frac{C}{2} \right) \\ &= \frac{c}{2} [a \cos B + a \cos A + b \cos B + b \cos A + a \cos B - a \cos A - b \cos B + b \cos A] \\ &\therefore (a - b)^2 \cos^2 \left(\frac{C}{2} \right) + (a + b)^2 \sin^2 \left(\frac{C}{2} \right) = \frac{c}{2} 2(a \cos B + b \cos A) \\ &\therefore (a - b)^2 \cos^2 \left(\frac{C}{2} \right) + (a + b)^2 \sin^2 \left(\frac{C}{2} \right) = c^2 \quad [\because c = a \cos B + b \cos A] \end{aligned}$$

Question 24: Find 'p' and 'q' if the equation

$px^2 - 8xy + 3y^2 + 14x + 2y + q = 0$ represents a pair of perpendicular lines.

$$z = 3x + 5y$$

Question 25: Maximize: $x + 4y \leq 24$, Subject to

$$3x + y \leq 21$$

$$x \geq 0, y \geq 0$$

$$x + y \leq 9,$$

Question 26: If $x = f(t)$ and $y = g(t)$ are differentiable functions of t , then prove that y is a

differentiable function of x and $\frac{dy}{dx} = \frac{\frac{dy}{dt}}{\frac{dx}{dt}}$, where $\frac{dx}{dt} \neq 0$

Hence find $\frac{dy}{dx}$ if $x = a \cos^2 t$ and $y = a \sin^2 t$

Question 27: $f(x) = (x-1)(x-2)(x-3)$, $x \in [0, 4]$ find 'c' if LMVT can be applied.

OR

Question 27: A rod of 108 meters long is bent to form a rectangle. Find its dimensions if the area is maximum.

Solution 27:

Perimeter of Rectangle = 108m

$$2 \times (\text{length} + \text{breadth}) = 108$$

$$\text{Length} + \text{Breath} = 54$$

$$l + b = 54$$

$$b = 54 - l \text{ -----(1)}$$

Area of Rectangle = Length $\times$ Breadth

$$A = l \times b$$

Now, we derivate the area,

$$\frac{dA}{dt} = \frac{d(i \times b)}{dt} \quad (\text{from 1})$$

$$= \frac{d(i \times (54 - i))}{dt}$$

$$= \frac{d(54l - l^2)}{dt}$$

$$= \frac{d(54l)}{dl} - \frac{d(l^2)}{dl}$$

$$= 54 - 2l$$

For area to be maximum, we put

$$\frac{dA}{dt} = 0$$

$$54 - 2l = 0$$

$$54 = 2l$$

$$l = \frac{54}{2}$$

$$l = 27\text{m}$$

$$b = 54 - 2l \text{ (from 1)}$$

$$= 27\text{m}$$

$$\frac{d^2 A}{dl^2} = \frac{d(54 - 2l)}{dl}$$

$$\frac{d^2 A}{dl^2} = \frac{d(54)}{dl} - \frac{d(2l)}{dl}$$

$$\frac{d^2 A}{dl^2} = 0 - 2$$

$$\frac{d^2 A}{dl^2} = -2 \leq 0$$

Therefore, area is maximum when length = 27m

Question 28: Prove that $\int \frac{dx}{\sqrt{x^2 + a^2}} = \log \left| x + \sqrt{x^2 + a^2} \right| + c$

Solution 28:

Put,

$$x = a \tan \theta$$

$$\therefore \tan \theta = \frac{x}{a}$$

Differentiating (i) w.r.t ' θ ', we get

$$dx = a \sec^2 \theta d\theta$$

$$\int \frac{dx}{\sqrt{x^2 + a^2}} = \int \frac{a \sec^2 \theta}{\sqrt{a^2 \tan^2 \theta + a^2}} d\theta$$

$$= \int \frac{a \sec^2 \theta}{\sqrt{a^2 (\tan^2 \theta + 1)}} d\theta$$

$$= \int \frac{a \sec^2 \theta}{\sqrt{a^2 \sec^2 \theta}} d\theta$$

$$= \int \frac{a \sec^2 \theta}{a \sec \theta} d\theta$$

$$= \int \sec \theta d\theta$$

$$= \log |\sec \theta + \tan \theta| + c_1$$

$$= \log \left| \sqrt{\sec^2 \theta} + \frac{x}{a} \right| + c_1$$

$$= \log \left| \sqrt{1 + \tan^2 \theta} + \frac{x}{a} \right| + c_1$$

$$= \log \left| \sqrt{1 + \frac{x^2}{a^2}} + \frac{x}{a} \right| + c_1$$

$$= \log \left| \frac{\sqrt{x^2 + a^2} + x}{a} \right| + c_1$$

$$= \log |\sqrt{x^2 + a^2} + x| - \log a + c_1$$

$$\therefore \int \frac{dx}{\sqrt{x^2 + a^2}} = \log |\sqrt{x^2 + a^2} + x| + c$$

Where,

$$c = -\log a + c_1$$

Question 29: Show that $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx = \frac{\pi}{8} \log 2$

Solution 29:

L.H.S,

$$I = \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$$

$$I = \int_0^{\frac{\pi}{4}} \log \left[1 + \tan \left(\frac{\pi}{4} - x \right) \right] dx \quad \left[\because \int_0^a f(x) dx = \int_0^a f(a-x) dx \right]$$

$$= \int_0^{\frac{\pi}{4}} \log \left[1 + \left(\frac{1 - \tan x}{1 + \tan x} \right) \right] dx$$

$$= \int_0^{\frac{\pi}{4}} \log \left(\frac{1 + \tan x + 1 - \tan x}{1 + \tan x} \right) dx$$

$$= \int_0^{\frac{\pi}{4}} \log \left(\frac{2}{1 + \tan x} \right) dx$$

$$= \int_0^{\frac{\pi}{4}} \log 2 dx - \int_0^{\frac{\pi}{4}} \log(1 + \tan x) dx$$

$$= \log 2 \int_0^{\frac{\pi}{4}} 1 dx - I$$

$$= \log 2 \cdot \left[x \right]_0^{\frac{\pi}{4}} - I$$

$$2I = \log 2 \cdot \left(\frac{\pi}{4} \right)$$

$$I = \frac{\pi}{8} \log 2$$

Hence, LHS = RHS proved.

Question 30: Solve the differential equation:

$$\frac{dy}{dx} + y \sec x = \tan x$$

Solution 30:

Given that,

$$\frac{dy}{dx} + y \sec x = \tan x$$

The equation is of the form

$$\frac{dy}{dx} + Py = Q$$

Where, $P = \sec x$ and $Q = \tan x$

$$\therefore I.F = e^{\int P dx} = e^{\int P dx} = e^{\log|\sec x + \tan x|}$$

$$= \sec x + \tan x$$

$\therefore$ Solution of the given equation is

$$y(I.F) = \int Q(I.F) dx + c$$

$$y(\sec x + \tan x) = \int \tan x(\sec x + \tan x) dx + c$$

$$= \int (\sec x \tan x + \tan^2 x) dx + c$$

$$= \sec x + \tan x - x + c$$

$$\therefore y(\sec x + \tan x) + (c - x)$$

$$\therefore y = 1 + \frac{c - x}{\sec x + \tan x}$$

OR

Question 30: Solve the differential equation;

$$(x + y) \frac{dy}{dx} = 1$$

Solution 30:

Given that,

$$(x + y) \frac{dy}{dx} = 1$$

Put,

$$\frac{dy}{dx} + Py = Q \text{ or } \frac{dx}{dy} + P_1x = Q_1$$

Dividing by $(x + y)$,

$$\frac{dy}{dx} = \frac{1}{(x+y)}$$

$$\frac{dx}{dy} = (x+y)$$

$$\frac{dx}{dy} - x = y$$

$$\frac{dx}{dy} + (-1)x = y$$

Find P_1 and Q_1

Comparing (1) with

$$\frac{dx}{dy} + P_1x = Q_1$$

$$P_1 = -1 \text{ and } Q_1 = y$$

So,

$$\begin{aligned} I.F &= e^{\int P_1 dy} \\ &= e^{\int (-1) dy} \\ &= e^{-y} \end{aligned}$$

Then, solution of the equation,

$$x \times I.F = \int Q_1 \times I.F . dy + c$$

Putting the values,

$$x \times e^{-y} = \int y \times e^{-y} . dy + c \text{-----(2)}$$

Let,

$$\begin{aligned}
 I &= \int y.e^{-y} dy \\
 &= y \int e^{-y} dy - \int \left[\frac{d}{dy} y \int e^{-y} dy \right] dy \\
 &= y \frac{e^{-y}}{-1} - \int 1. \frac{e^{-y}}{-1} dy \\
 &= -y.e^{-y} + \int e^{-y} dy \\
 &= -y.e^{-y} + \frac{e^{-y}}{-1} \\
 &= -y.e^{-y} - e^{-y}
 \end{aligned}$$

Putting the value of I in equation (2)

$$\begin{aligned}
 xe^{-y} &= \int y \times e^{-y}.dy + C \\
 xe^{-y} &= -ye^{-y} - e^{-y} + C
 \end{aligned}$$

Dividing by e^{-y}

$$\begin{aligned}
 x &= -y - 1 + Ce^y \\
 x + y + 1 &= Ce^y
 \end{aligned}$$